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EDA Toolkit: A Lightweight Python Library for Exploratory Data Analysis and Reporting

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Software

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Summary

EDA Toolkit is a lightweight Python package for conducting and visualizing exploratory data analysis (EDA). It provides flexible plotting tools, profiling summaries, and exportable Excel reports tailored for both academic and industry workflows. Designed to be highly customizable and easily integrated into notebooks or pipelines, EDA Toolkit helps users rapidly understand and communicate data characteristics.

Statement of need

Exploratory Data Analysis (EDA) is a critical step in any data science project. It helps researchers understand the structure of a dataset, detect data quality issues, identify meaningful patterns, and shape the direction of analysis. While tools such as `pandas_profiling` and `sweetviz` provide automated reports, they often lack the flexibility, transparency, and formatting needed in professional or academic settings.

EDA Toolkit addresses these limitations by offering a modular and customizable set of tools designed for clarity, reproducibility, and high-quality presentation. It supports both academic research and applied data science use cases, with an emphasis on producing outputs that are publication-ready and easy to interpret.

More than several examples throughout the documentation and figures in this paper are based on the Adult Income dataset from the UCI Machine Learning Repository (Dua & Graff, 2019; Kohavi, 1996). This real-world tabular dataset offers a rich foundation for testing demographic segmentation, group comparisons, and reporting tools, particularly in contexts involving income classification and social variables. It serves as a practical benchmark for demonstrating the capabilities of functions such as `generate_table1()` and outlier visualization utilities within the EDA Toolkit.

You can install `eda_toolkit` directly from PyPI:

```
pip install eda_toolkit
```

Source code: https://github.com/lshpaner/eda_toolkit
Documentation: https://lshpaner.github.io/eda_toolkit_docs
DOI: https://doi.org/10.5281/zenodo.13162633

Income Distribution by Age Group

To explore income distribution patterns across age groups, we used the `stacked_crosstabs()` function. This function generates both raw and normalized stacked bar charts, allowing for

quick visual comparison of categorical variable relationships.

The top panel of the figure displays the count of individuals in each age group, broken down by income level (<=50K vs >50K). Younger age groups (18–29 and 30–39) dominate in population size, but a higher proportion of individuals earning over $50K is observed in the 40–59 range.

The bottom panel presents a normalized version of the chart, where each bar sums to 1. This highlights relative prevalence: while most younger individuals earn less than $50K, the proportion of higher earners increases notably in the 30–59 age range. The graph helps identify trends not just in absolute counts but in income composition across demographic segments.

Figure 1: Stacked Bar Graphs of Income by Age Group

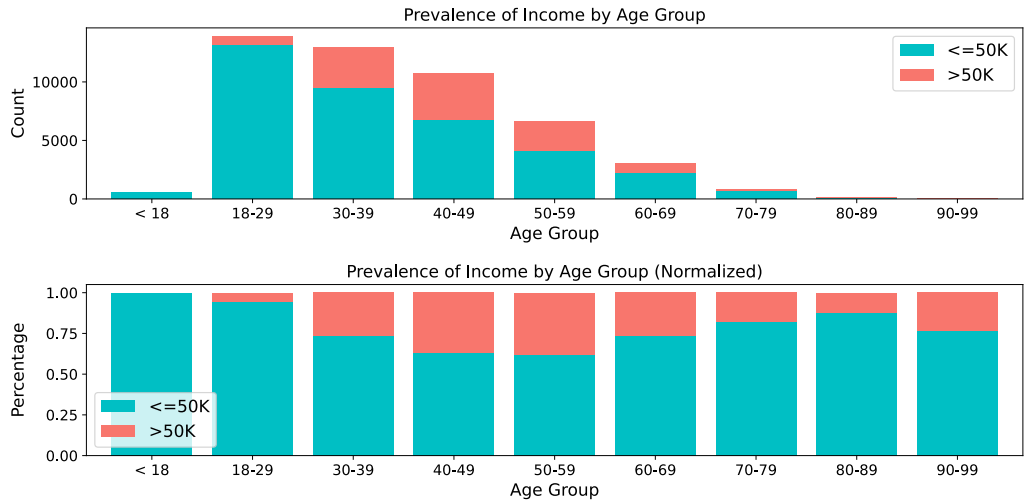

Figure 1: Figure 1

Table 1 Generation

The `generate_table1()` function allows users to produce clean, formatted descriptive tables often used in clinical and academic research. The output includes summaries by group and supports filtering by data type, making it easier to communicate sample characteristics without relying on external tools like Excel.

Table 1: Group-wise descriptive statistics using the `generate_table1()` function on the UCI Adult Income dataset.

Variable	Count	Proportion (%)	<=50K (n = 37,155)	>50K (n = 11,687)	P-value
fnlwgt_bin	48,842	100.00	37,155	11,687	0.66
fnlwgt_bin = Bin 1	42,729	87.48	32,517 (87.52%)	10,212 (87.38%)	
fnlwgt_bin = Bin 2	5,898	12.08	4,466 (12.02%)	1,432 (12.25%)	
fnlwgt_bin = Bin 3	186	0.38	148 (0.40%)	38 (0.33%)	
fnlwgt_bin = Bin 4	22	0.05	18 (0.05%)	4 (0.03%)	
fnlwgt_bin = Bin 5	7	0.01	6 (0.02%)	1 (0.01%)	

Variable	Count	Proportion (%)	<=50K (n = 37,155)	>50K (n = 11,687)	P-value
age_group	48,842	100.00	37,155	11,687	0.00
age_group = 18-29	13,920	28.50	13,174 (35.46%)	746 (6.38%)	
age_group = 30-39	12,929	26.47	9,468 (25.48%)	3,461 (29.61%)	
age_group = 40-49	10,724	21.96	6,738 (18.13%)	3,986 (34.11%)	
age_group = 50-59	6,619	13.55	4,110 (11.06%)	2,509 (21.47%)	
age_group = 60-69	3,054	6.25	2,245 (6.04%)	809 (6.92%)	
age_group = 70-79	815	1.67	668 (1.80%)	147 (1.26%)	
age_group = < 18	595	1.22	595 (1.60%)	0 (0.00%)	
age_group = 80-89	131	0.27	115 (0.31%)	16 (0.14%)	
age_group = 90-99	55	0.11	42 (0.11%)	13 (0.11%)	
age_group = 100 +	0	0.00	0 (0.00%)	0 (0.00%)	

Outlier and anomaly detection support

The library includes functions to identify and visualize outliers based on distributional thresholds or robust statistics. This capability supports detecting data quality issues early, understanding variable spreads, and guiding preprocessing decisions.

When we examine the age distribution using the `kde_distributions()`, the distribution of the age variable is visibly right-skewed, with the kernel density estimate (KDE) curve exhibiting a longer tail on the upper end of the age spectrum. This indicates that the dataset contains a larger proportion of individuals in the younger to middle-aged range, while older individuals are less frequent. The mean age, represented by a blue dashed line, is slightly greater than the median (black dashed line), further confirming the right-skewed nature of the data. Most of the data fall within the $\pm$ 1 standard deviation interval (purple dashed lines), which ranges approximately from 25 to 55 years. The $\pm$ 2 and $\pm$ 3 standard deviation bands (green and gray lines, respectively) extend out to include nearly the full age range, but observations beyond 70 or below 10 are uncommon. The presence of skewness and a small number of extreme values justifies the use of a Box-Cox transformation, which was applied in subsequent analysis to normalize the distribution and support downstream modeling assumptions.

Figure 2: Distribution of Age

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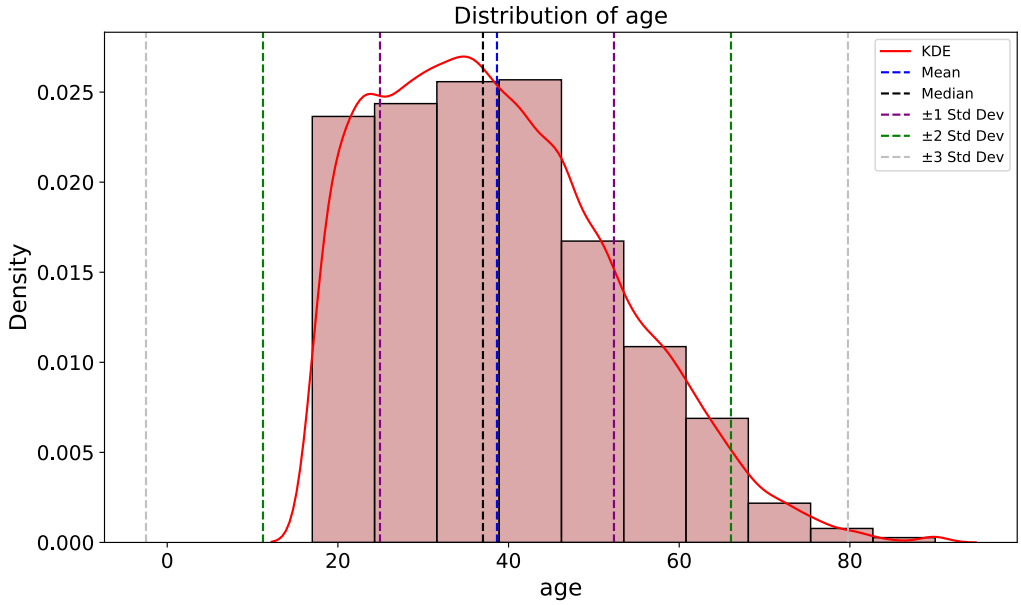

Figure 2: Figure 2

Box-Cox Transformation

To correct for right skewness and enhance model interpretability, we applied the Box-Cox transformation to the age variable using the `data_doctor()` function. This transformation is designed to normalize positively skewed continuous variables by applying a power transformation governed by a parameter λ, which was empirically estimated at **0.1748** for this variable.

The Box-Cox transformation is defined as:

$$y(\lambda) = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \ln(y), & \text{if } \lambda = 0 \end{cases}$$

Where:
- $y(\lambda)$ is the transformed value
- (y) is the original, strictly positive continuous variable
- (λ) is the transformation parameter selected to best approximate normality

The transformed `age_boxcox` variable exhibits improved symmetry and reduced kurtosis, making it more suitable for statistical modeling and visualization.

This transformation was applied using the `data_doctor()` function from the `eda_toolkit` library, which also produced visual diagnostics (KDE, histogram, and boxplot), computed summary statistics, and added a new column `age_boxcox` to preserve the transformed values.

The function outputs distribution plots (KDE, histogram, boxplot) on the transformed scale, computes summary statistics, and calculates the interquartile range (IQR) for potential outlier detection. For this analysis, cutoff filtering was disabled, but the lower and upper bounds are shown for reference.

A new column `age_boxcox` was added to the DataFrame to retain the transformed values without overwriting the original data.

```
DATA DOCTOR SUMMARY REPORT
```

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```
+------------------------------+--------------------+
| Feature                      | age                |
+------------------------------+--------------------+
| Statistic                    | Value              |
+------------------------------+--------------------+
| Min                          |             3.6664 |
| Max                          |             6.8409 |
| Mean                         |             5.0163 |
| Median                       |             5.0333 |
| Std Dev                      |             0.6761 |
+------------------------------+--------------------+
| Quartile                     | Value              |
+------------------------------+--------------------+
| Q1 (25%)                     |             4.5219 |
| Q2 (50% = Median)            |             5.0333 |
| Q3 (75%)                     |             5.5338 |
| IQR                          |             1.0119 |
+------------------------------+--------------------+
| Outlier Bound                | Value              |
+------------------------------+--------------------+
| Lower Bound                  |             3.0040 |
| Upper Bound                  |             7.0517 |
+------------------------------+--------------------+

New Column Name: age_boxcox
Box-Cox Lambda: 0.1748
```

Figure 3: Box-Cox Transformed Age

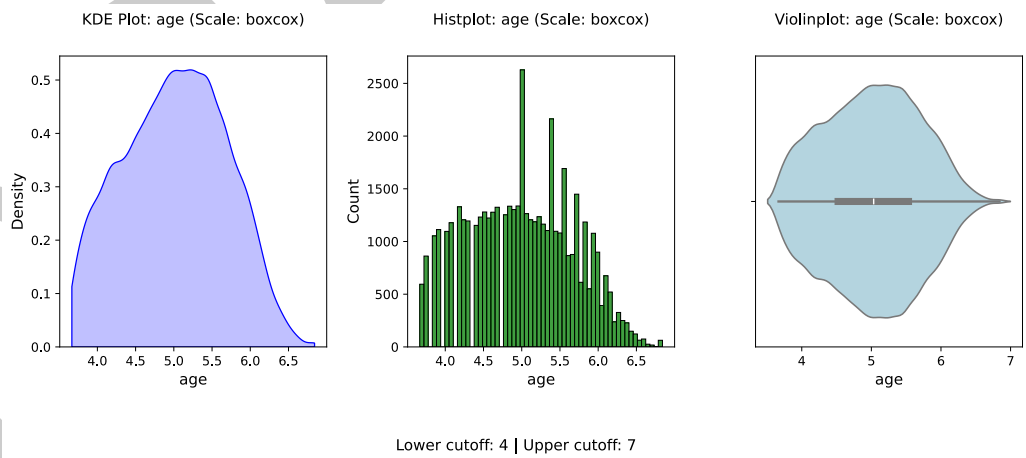

Figure 3: Figure 2

Acknowledgements

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References

Dua, D., & Graff, C. (2019). *Adult data set*. https://archive.ics.uci.edu/ml/datasets/adult

Kohavi, R. (1996). *Census income*. UCI Machine Learning Repository. https://doi.org/10.24432/C5GP7S

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